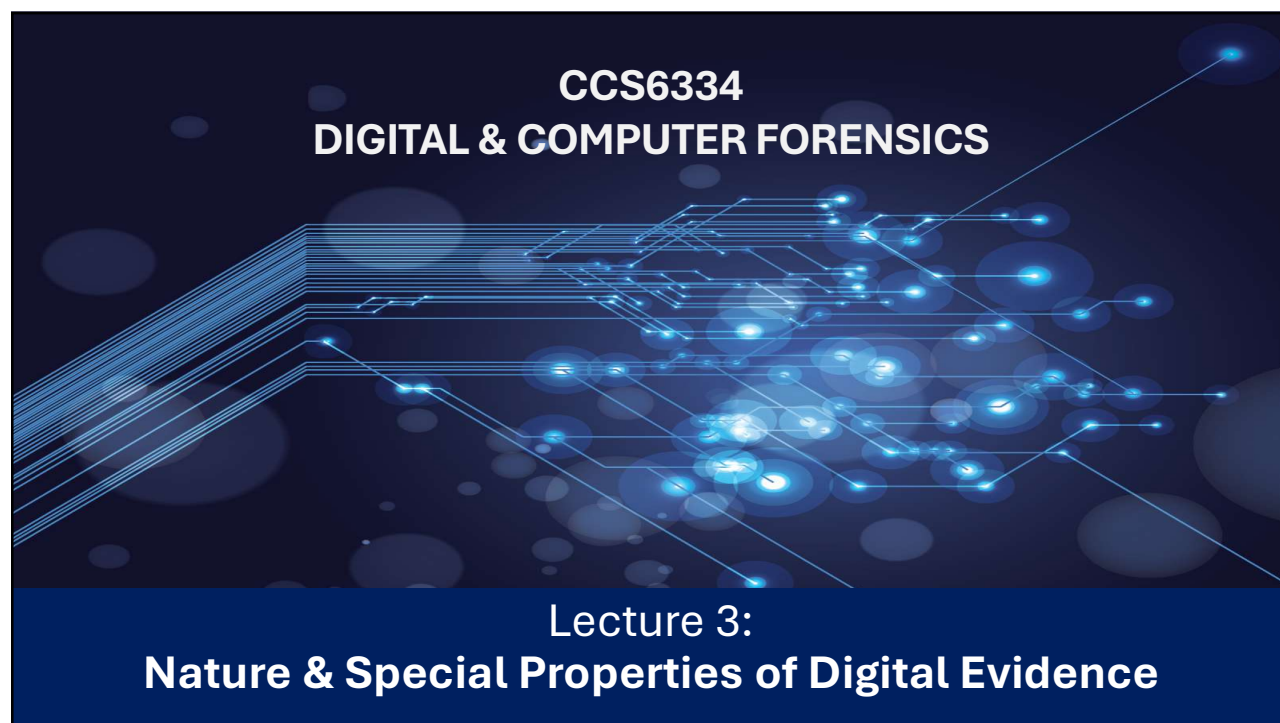




1

Learning Objectives

After studying this chapter, you should be able to:

- Defining digital evidence and its use
- Explain properties of digital evidence
- Highlight the technical complexities of digital evidence and challenges to sound analysis
- Explain the requirements for determining the admissibility of digital evidence

2

DIGITAL EVIDENCE



3

Defining Digital Evidence



- **Digital Evidence:** Information in digital form found on any device containing a microchip or processed/stored digitally (e.g., computers, phones, IoT devices).
 - It is revealed during the forensics investigation while examining storage media, monitoring network traffic, or making duplicate copies of digital data
- **Nature:** Data represented numerically in binary, unlike analog electronic data.
- **Example Case:**
 - An Australian narcotics trafficker hid a spreadsheet of transactions inside a microchip implanted in a dog.
 - Discovered and recovered as digital evidence.
- **Legal Context:**
 - Classified as real or documentary evidence in court.
 - Includes emails, documents, spreadsheets, databases, system logs, audio, images, and videos.
 - Often interpreted and presented through expert testimony.

4

The log file shows the period of imaging of a 250-GB drive

```
2015-11-26 11:58:26 syslogd started: BusyBox v1.16.2
2015-11-26 11:58:26 kernel: Initializing cgroup subsys cpuset
2015-11-26 11:58:26 kernel: Initializing cgroup subsys cpu
2015-11-26 11:58:26 kernel: Linux version 3.4.49-x86_eriX (andersen@git.perlustr.com) (gcc version 4.7.2 (GCC) ) #1 SMP Fri Jun 14 17:06:52 MDT 2013

2015-11-26 11:58:27 root: System Information
2015-11-26 11:58:27 root: Product Name: HP Compaq 4000 Pro SFF PC
2015-11-26 11:58:27 root: Version:
2015-11-26 11:58:27 root: UUID: XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
2015-11-26 11:58:27 root: SKU Number: LE123PARAB0
2015-11-26 11:58:27 root: Family: 103C_53307F G+D
2015-11-26 11:58:27 root: Manufacturer: Hewlett-Packard
2015-11-26 11:58:27 root: Version: Not Specified
2015-11-26 11:58:27 root: Chassis Information
2015-11-26 11:58:27 root: Manufacturer: Hewlett-Packard
2015-11-26 11:58:27 root: Lock: Not Present
2015-11-26 11:58:27 root: Serial Number: XXXXXXXX
2015-11-26 11:58:27 root: Boot-up State: Safe
2015-11-26 11:58:27 root: Power Supply State: Safe
2015-11-26 11:58:27 root: Security Status: Unknown

2015-11-26 12:01:09 imager: Making a Image of /dev/sda
2015-11-26 12:01:09 imager: A 250.1 GB SAMSUNG HD256GJ Hard Drive
2015-11-26 12:01:09 imager: Image will be stored on /dev/sdc
2015-11-26 12:01:09 imager: A 2.000 TB Seagate Expansion Hard Drive
2015-11-26 12:01:09 imager: Output File Format: Ilook Default Image Format
2015-11-26 12:01:09 imager: Output File Size: Unlimited
2015-11-26 12:01:09 imager: Compression: Enabled
2015-11-26 12:01:09 imager: Encryption: Disabled
2015-11-26 12:01:09 imager: Case Number: XXXXXXXXXXXXXXXX
2015-11-26 12:01:09 imager: Agent Name: Richard Boddington
2015-11-26 12:01:09 imager: Machine Owner: XXXXXXXX
2015-11-26 12:01:09 imager: Seizure Address: XXXXXX Perth
2015-11-26 12:01:09 imager: Known Passwords: XXXXXX
2015-11-26 12:01:14 imager: User exited the Final Options Menu
2015-11-26 12:01:14 imager: Beginning Image operation
2015-11-26 12:01:14 kernel: ntfs info: NTFS volume version 3.1 (cluster_size 32768, PAGE_CACHE_SIZE 4096).
2015-11-26 12:01:14 imager: Opened output file "/IlookImager/Ilook.001/XXXXXXXX001.asb"
2015-11-26 12:01:14 imager: Calibrating "/dev/sdc" for output, a 2.000 TB NTFS Filesystem on USB0
2015-11-26 12:01:16 imager: Image is being stored to /Ilook.001/Ilook.001/XXXXXXXX001.asb
2015-11-26 12:01:16 imager: A 2.000 TB NTFS Filesystem on USB0
2015-11-26 12:01:16 imager: Image is being stored to /Ilook.001/XXXXXXXX001.asb

2015-11-26 13:00:08 imager: Image Complete
2015-11-26 13:00:08 imager: Image was completed successfully.
2015-11-26 13:00:08 imager: Read : 250.1 GB (250059350016 bytes)
2015-11-26 13:00:08 imager: Written : 72.74 GB (72744460583 bytes)
2015-11-26 13:00:08 imager: Total Processed: 250.1 GB (250059350016 bytes)
2015-11-26 13:00:08 imager: Image Speed : 70.80 MB/sec
2015-11-26 13:00:08 imager: Elapsed Time : 0h 58m 52s
2015-11-26 13:00:08 imager: Compression : 70.91 percent
2015-11-26 13:00:08 imager: Bad Sectors : 0
2015-11-26 13:00:08 imager: Copying logfile to Ilook.001/
2015-11-26 13:00:08 imager: Creating metadata
```

View of a text file using IlookIX

Find and Replace

Find:

Search: All

☐ Match case

☐ Find whole words only

☐ Regular expression

Find Next Cancel

File List (2)

Viewer

LABORATORY CHEMICALS THAT ARE EXPLOSIVE
OR MAY DEVELOPE EXPLOSIVE POTENTIAL ON DETERIORATION

1 Explosive Laboratory Chemicals

The following list of explosive chemicals that may occur in the laboratory is drawn from the *Australian Code for the Transport of Dangerous Goods by Road and Rail* and from Appendix C4 (Explosive chemicals) of M.I. Pitt and E. Pitt *Handbook of Laboratory Waste Disposal*. The list may not be complete.

Appropriate care should be exercised in the storage and handling of these materials.

acetyl peroxide

Viewer Properties Hex View Plain Text View Log Histogram Comments

5

The Properties sheet of a file viewed using IlookIX

Properties

Property	Value
File	WelcomeFax.tif
Top Level Parent	Maximal No VSS (0)
Key	17422 / 100.0.5247.17316
Barcode	
Type	tif
Path	
Full Path	Maximal No...
Created	Tuesday, April 12, 2011 8:17:28 AM
Last Accessed	Wednesday, September 11, 2013 8:10:58 PM
Last Modified	Tuesday, April 12, 2011 8:17:28 AM
Additional date records	
Virus/Trojan scanned	No
Truncated	No
Attributes	A---
Metadata	[Exif IFD0] Unknown tag (0x00fe): 0 Unknown tag (0x0100): 816 Unknown tag (0x0101): 1056 Unknown tag (0x0102): 8 8 8 Unknown tag (0x0103): 5 Unknown tag (0x0106): 2 Unknown tag (0x0111): [212 System.Int32s] Unknown tag (0x0115): 3 Unknown tag (0x0116): 5 Unknown tag (0x0117): [212 System.Int32s] X Resolution: 96/1 dots per inch Y Resolution: 96/1 dots per inch Unknown tag (0x011c): 1 Resolution Unit: Inch Unknown tag (0x013d): 1 No thumbnail data exists in this image
Hardlinks (1)	
Linked Object Key	17423
Full Path	Maximal No VSS:0:\ProgramData\Microsoft\Windows NT\MSFax\VirtualInbox\en-US\WelcomeFax
ACL Data	0

Viewer Properties Hex View Plain Text View Log Histogram Comments

File categories based on search terms for later analysis

ISeekExplorer Explorer

Categories	Description
Available FileTypes	39,629 File/e-mail load categorized by type
Case categories	0 Evidential material categories
Search Results	3 Previous search results
fraud: 26/06/2015 8:14:19 PM	566 fraud
hutchens: 26/06/2015 7:24:...	67 hutchens
hutchens: 26/06/2015 7:32:...	1,422 hutchens

6

3



Types of Evidence

- There are 4 general categories
- Real – physical, tangible evidence such as an object, a tape recording, a printout, photograph, etc. that the court can examine for itself
 - Documentary – typically, any type of document or facsimile that records details of an event including contracts, logs, records, diaries, photos, carvings, maps etc.
 - Demonstrative – type of representation of an object that represents, illustrates, or reenacts something else, such as injuries, physical characteristics, or events
 - Testimonial – verbal statements given by witnesses made under oath, also known as direct evidence
- NOTE: There are also other ways to categorize, this is just one way!

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REAL	DOCUMENTARY	TESTIMONIAL	DEMONSTRATIVE
Biological samples  Blood  Saliva  Hair  Skin cell Ballistic evidence 	Paper trail  Documents 	Eyewitness  Expert testimony 	Simulation/animation  Maps/Graphs/Chart 

8



The Use of Digital Evidence

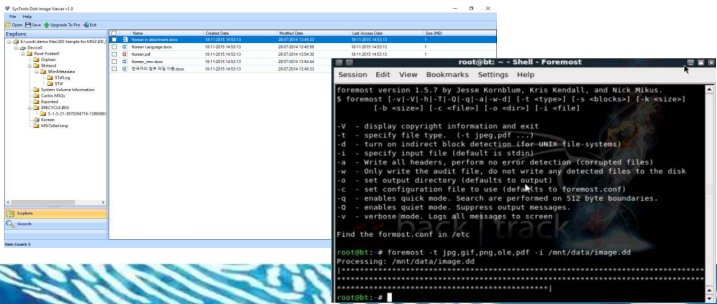
- **Purpose:**
 - Used to prove or refute disputed facts in legal cases.
 - Supports investigations by linking the crime, victim, and perpetrator.
 - Must be reliable to sustain or challenge legal hypotheses.
- **Forms and Sources:**
 - Recovered from computers, servers, and digital devices.
 - Can serve in both criminal and civil cases — sometimes the only evidence available.
- **Collection & Preservation:**
 - Evidence is obtained using forensic imaging (bit-for-bit copy).
 - Write blockers prevent contamination of the original data.
 - Forensic image formats: .AFF, .ASB, .E01, .dd, .VMDK, .VDI.
- **Benefits:**
 - Enables exact replication without altering the original device.
 - Preserves integrity of evidence for court presentation.
 - Reduces risk of data loss or contamination.

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Tools to Collect Digital Evidence

- Special forensics tools are required to open and examine the data held in an image, and they do so without altering the image and contaminating the evidence, in the tradition of sound forensic practice
- It allows files of importance to be extracted and displayed using the forensic software, but it does not change the composition and integrity of the image or the files and metadata it contains
- Looking at system files, navigating file directories, and opening cache folders to view browsing history are now supported by enhanced forensic software.
- Commercial tools like *Disk Image Viewer* or open-source tools like *Foremost* allow extraction of files from disk copies

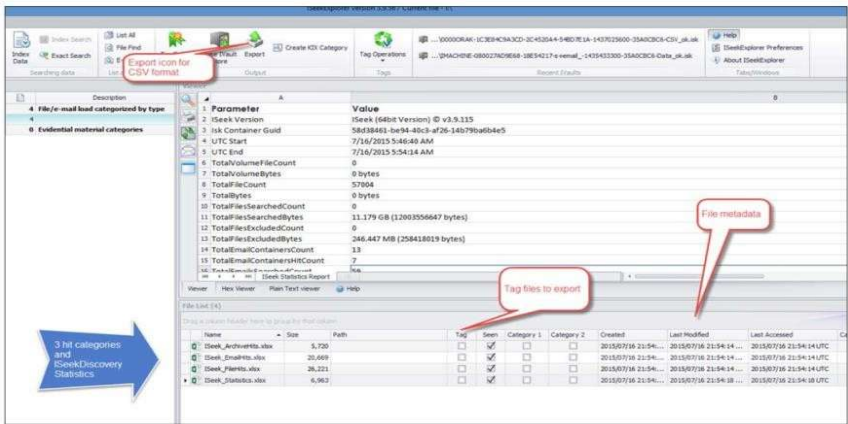


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Tools to Collect Digital Evidence (Cont.)



- Recovering and viewing information from network server computers may be difficult/restricted because the organization may not want to stop its normal business, because each server or folder on a server might need to be imaged (i.e. taken offline)



The ISeekDiscovery automaton: search term hit statistics and corresponding recovered files

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Tools to Collect Digital Evidence (Cont.)



- Mobile phones and other handheld devices are not imaged in the same way as desktop because of different filesystem, hardware architecture, encryption, etc.
- Some devices are particularly difficult to image and extract data from, i.e. iOS devices, but tools from developers like Cellebrite and Grayshift show it is possible and thus we have the perpetual cat-and-mouse game between the parties involved



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Tools to Collect Digital Evidence (Cont.)



- For **mobile devices**, extra care must be taken to ensure that it does not connect to a mobile network, local Wi-Fi, or Bluetooth devices in the proximity
- SIM cards are removed and may be examined separately to protect contamination to the phone by inadvertent local connections
- This is to ensure the possibility of remote execution of operations like device wipe or factory reset is not possible
- Remote-wiping applications, intended to allow phone owners to wipe private data on their phones in the event of theft or loss, are now used by those wishing to frustrate investigators from accessing potential evidence
- Mobile devices are fast becoming the tool of choice for communication and thus deserves a topic/chapter on its own (to be covered later)

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Mobile Device Forensics: Key Points



- **Different from computer imaging** due to unique hardware, interfaces, and strong encryption (e.g., iPhone encrypted sectors).
- **Access limitations:** Physical extraction no longer possible on iPhone 4S+ and many newer Androids; often limited to logical extraction or backups.
- **Specialized tools:** Software such as **Cellebrite** and **XRY** performs data extraction, creates proprietary evidence containers, and generates logs for analysis.
- **Data sources:** Phone memory, device characteristics, SIM cards, and SD cards.
- **Risk of remote wiping:** Devices must be isolated from mobile networks, Wi-Fi, and Bluetooth to prevent data loss.
- **Write-blocking challenges:** Mobile devices cannot be fully write-blocked; small programs must be installed, creating unavoidable risk of minor data alteration.

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DIGITAL EVIDENCE CHARACTERISTICS



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What are the Characteristics of Digital Evidence?



- The main characteristics of digital evidence is its relevance and weight (influence)
- **Relevance** - the connection between digital evidence and the fact(s) that is/are to be proved
- **Weight** – how much the digital evidence changes probability of the fact.

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What are the Characteristics of Digital Evidence?



ALL EVIDENCE MUST BE:

- **Admissible** – related to the fact being approved
- **Authentic** – real and related to the incident in a proper way
- **Complete** – prove guilt or innocence
- **Reliable** – no doubt about the authenticity or veracity of evidence
- **Believable** – clear and understandable

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Characteristics of Digital Evidence & Legal Context



- Courts distinguish direct (witness) evidence and indirect (circumstantial) evidence.
- Human testimony must come from first-hand sensory observation; hearsay is generally inadmissible.
- Digital evidence is considered circumstantial and often treated like documentary evidence.
- Its admissibility depends on reliability, accuracy, and whether records are created in the regular course of business.
- Courts increasingly accept digital evidence, especially with supporting expert explanation

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Nature of Digital Evidence & Forensic Interpretation



- Digital evidence is indirect, lacks a “voice,” and requires expert interpretation to infer meaning.
- It is probabilistic, requiring correlation with other evidence for stronger credibility.
- Forensic practitioners explain technical findings, context, and inferences to assist the court.
- Ideally, digital evidence corroborates or is corroborated by other evidence for stronger admissibility.
- US Federal Rules of Evidence allow certain state-of-mind statements (e.g., emails, social media posts).

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Important Forensics Principles



- **ACPO** Principles of Digital Evidence (4)
 - No action taken should change data
 - ONLY competent people access original data
 - CHAIN OF CUSTODY AT ALL TIMES !!!
 - Person in charge is accountable
- Scientific Working Group on Digital Evidence (SWGDE) Principles, Standards & Criteria

ACPO = Association of Chief Police Officers

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Collecting Physical Evidence



Physical evidence includes:

- Removable media
- Cables
- Publications
- All computer equipment, including peripherals
- Item taken from trash
- Miscellaneous items

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Collecting Physical Evidence (Cont.)



Tag all the object identified as evidence and mention all the required details on the tag, such as the time, date, incident responder's name, and control number.

Collection evidence from:

- Powered On/ Powered Off computers
- Network connected computers
- Open files and startup files

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What is Chain of Custody?



- Demonstrate the progression of evidence as it travels from the original evidence location to the forensic laboratory
- Covers the collection, handling, storage, testing, and disposition of evidence
- Documentation should list all the people involved in the collection and preservation of evidence and their actions.

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What are in the Evidence Bag Content List?



The panel on the front of evidence bags must, at the very least, contain the following details:

- Date and time of seizure
- Incident responder who seized the evidence
- Exhibit number
- Where the evidence was seized from
- Details of the contents of the evidence bag
- Submitting agency and its address

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DIGITAL EVIDENCE COMPLEXITIES



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Technical Complexities of Digital Evidence



1. Complexity & Expertise Required

- Digital evidence is highly technical and requires specialized skills to interpret.
- Metadata and file information are useful, but understanding their nature is essential.
- Digital evidence often forms *indirect evidence*, making interpretation more challenging.

2. Challenges in Authenticity & Attribution

- Over 90% of documents are now digital → easier to alter, harder to detect tampering.
- Linking suspects to events through metadata alone is difficult and sometimes incomplete.
- Evidence may be missing, erased, or fragmented, breaking the chain of evidence.

3. Anonymous Attacks & Jurisdiction Issues

- Remote attacks complicate linking offenders to actions.
- Attackers may operate from countries that offer safe havens.
- Extradition and international cooperation often slow down investigations.

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Malleability of Digital Evidence & Metadata Limitations



1. Digital Evidence Is Easily Altered

- Improper handling can modify or destroy evidence (e.g., booting a device, shutdown erasing RAM).
- Forensic tools must minimize contamination during acquisition.
- Deleted or overwritten data may leave minimal recoverable traces.

2. Malleability & Hidden Artifacts

- Evidence can be intentionally removed; recovery requires advanced expertise and sometimes luck.
- Artifacts like *thumbcache* databases may reveal previously existing files.

3. Metadata Can Be Misleading

- Metadata records creation/modified/access times but not all activities.
- “Last accessed” timestamps can be changed by antivirus scans, not just user actions.
- Word documents may store previous versions, but only if versioning is enabled.



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Malleability of Digital Evidence & Metadata Limitations



Creation date	Last modified date	Last accessed date
10/10/2012 20:50:32	10/10/2012 20:50:32	10/10/2012 20:50:32

File metadata timestamps

Creation date	Last modified date	Last accessed date
10/10/2012 20:50:32	10/10/2012 20:50:32	15/10/2012 18:03:49

Last accessed timestamp altered

Creation date	Last modified date	Last accessed date
10/10/2012 20:50:32	16/10/2012 11:50:59	16/10/2012 11:50:59

Last modified and accessed dates



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Recovering Files from Unallocated Space (Data Carving)



- Deleted files often leave fragments in the **unallocated space** of a hard drive.
- Because filesystem allocation data is no longer available, these fragments cannot be located through normal file system structures.
- Traditionally, investigators used manual hex-editor analysis to identify file headers, footers, and internal patterns, an intensive and highly technical process.
- Despite modern automation, file carving remains essential, especially in cases where recovering deleted or partially overwritten files is critical to an investigation.

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Date and Time Problems in Digital Evidence



- Timestamps are critical** for reconstructing events, but device clocks are often unreliable.
- Computer clocks can drift, run fast/slow, or lack synchronization with trusted time sources (e.g., NIST), making logged events inaccurate.
- Users may **manually change time or time zones**, especially when travelling; Windows Registry records these adjustments.
- Different filesystems store timestamps differently (UTC vs. local time), requiring awareness of the **origin time zone** during analysis.
- Windows Registry may display timestamps in **UTC**, even when the system shows local time, analysts must convert correctly to avoid misinterpretation.

Name	0	Type	Size	Last Modified
usb_vid_054c&pid...		REG_SZ	18	03 Jan 2016 04:52:51
usb_vid_054c&pid...		REG_SZ	40	03 Jan 2016 04:52:51
usb_vid_054c&pid...		REG_SZ	46	03 Jan 2016 04:52:51
usb_vid_054c&pid...		REG_SZ	66	03 Jan 2016 04:52:51
usb_vid_054c&pid...		REG_SZ	60	03 Jan 2016 04:52:51
usb_vid_054c&pid...		REG_SZ	32	03 Jan 2016 04:52:51
usb_vid_054c&pid...		REG_SZ	30	03 Jan 2016 04:52:51
usb_vid_054c&pid...		REG_SZ	64	03 Jan 2016 04:52:51
usb_vid_054c&pid...		REG_SZ	70	03 Jan 2016 04:52:51
usb_vid_054c&pid...		REG_SZ	70	03 Jan 2016 04:52:51
usb_vid_054c&pid...		REG_SZ	50	03 Jan 2016 04:52:51
usb_vid_054c&pid...		REG_SZ	70	03 Jan 2016 04:52:51
usb_vid_054c&pid...		REG_SZ	70	03 Jan 2016 04:52:51
usb_vid_054c&pid...		REG_SZ	52	03 Jan 2016 04:52:51
usb_vid_054c&pid...		REG_SZ	64	03 Jan 2016 04:52:51

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DIGITAL EVIDENCE VALUE & ADMISSIBILITY



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Determining the Value & Admissibility of Digital Evidence



- Digital evidence must have probative value, contributing meaningfully to the chain of evidence in a case.
- To be admitted in court, it must meet mandatory legal conditions; failure to do so renders it inadmissible.
- Admissibility is governed by legislation and common law, with some jurisdictions (e.g., USA) giving judges broad discretion.
- Forensic expertise is essential to properly identify, analyze, and justify the admissibility of digital evidence—often insufficiently challenged in court.
- Courts increasingly scrutinize digital evidence due to complexities and potential disputes affecting its reliability.
- Legal systems differ:
 - Inquisitorial systems (France, Germany, many parts of Africa, South America, Asia) focus on truth through thorough investigation.
 - Adversarial systems (UK, Australia, North America) rely on competition between prosecution and defense, which may obscure truth.

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Evidentiary Weight of Digital Evidence



- Evidence has value when it is plausible, relevant, and contributes to determining a suspect's identity or culpability.
- Judges and juries decide the weight of evidence, not practitioners.
- Practitioners must assess the relevance and significance of digital artifacts they discover, often being the first to identify them.
- Investigators typically work toward a specific brief (e.g., fraud), guiding what types of digital evidence support the case.
- Legal debate may arise over the strength of inferences, especially if evidence risks misinterpretation or appears unreliable.
- Strong evidentiary weight requires sufficient, corroborated evidence to support or refute elements of the case.
- Evidence that becomes inadmissible weakens the case and may cause it to collapse; many cases never reach trial due to insufficient evidentiary support.

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Understanding the Admissibility of Digital Evidence



- Disputes over admissibility are common, yet digital evidence is often unchallenged due to its technical complexity and novelty.
- Lawyers may lack understanding of digital evidence, leaving judges or juries to evaluate admissibility.
- Practitioners advise investigative and legal teams on whether evidence is lawfully acquired, relevant, and reliable.
- Complex cases may involve voir dire hearings ("trial within a trial") where experts present technical arguments to determine admissibility.
- Weight of evidence is separate from admissibility: evidence must comply with court guidelines to be legally tendered.
- Admissible digital evidence must be:
 - Lawfully acquired
 - Relevant to proving or disproving aspects of the case
 - Reliable and technically sound

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Lawful Acquisition of Digital Evidence



- Evidence must be obtained lawfully to be admissible in court.
- Lawful acquisition may occur via:
 - Owner consent (written permission for access or imaging)
 - Statutory powers allowing seizure by law enforcement without a warrant
 - Search and seizure warrants specifying location, persons, and items to be seized
- Practitioners must understand legal requirements for seizing devices (computers, mobile phones, networks) to ensure evidence remains admissible.
- Recent rulings (e.g., US Supreme Court) require separate warrants for mobile phones, with strict timelines and procedural compliance.
- Exceeding warrant scope or seizing evidence without proper authorization may lead to suppression of evidence, though untainted supporting evidence may still be admissible.
- Proper training, policies, and technology are essential for effective and lawful digital evidence collection.

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Importance of Relevance in Digital Evidence



- Digital evidence must be **relevant, non-hearsay, and not overly prejudicial** to be admissible.
- Courts often require practitioners to **explain the creation and storage** of digital evidence to clarify its relevance.
- **Judges and juries** assess both the relevance and the plausibility of evidence when determining its weight.
- Misinterpretation or overinterpretation by inexperienced practitioners can affect admissibility or perceived weight.
- Digital evidence undergoes a **logical relevance test**, similar to traditional evidence, ensuring it contributes meaningfully to the case without being treated as inherently mysterious or special.

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Reliability of Digital Evidence



- Reliability is generally defined by **authenticity, accuracy, and fidelity** → evidence must be what it purports to be.
- Digital evidence must remain **untainted and uncontaminated** throughout collection and analysis.
- Key factors affecting reliability:
 - **Forensic tools** used in collection and preservation
 - **Recovery and protection processes** ensuring data integrity
 - **Absence of human or machine contamination**
 - **Device and network security** protecting the evidence

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Importance of Reliability of Forensic Tools & Processes



- Practitioners use forensic tools to search, recover, and analyze large datasets and complex file structures.
- Evidence contamination can occur during acquisition or processing, potentially destroying or altering important data.
- Forensic best practice: create forensic images to preserve original evidence for further analysis.
- Faulty tools or processes can result in data loss, overwriting, or false information, jeopardizing case integrity.
- Courts may question admissibility if evidence is suspected of contamination; reliability is critical for legal acceptance.
- Daubert Test (US) evaluates the validity of forensic tools and processes; similar standards are essential globally.
- Reliable tools and processes ensure evidence remains trustworthy, authentic, and admissible.

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Evaluating Computer/Network Evidence Preservation



- Organizations rely on computer-based systems and networks to store and manage critical information.
- Networked databases face heightened risks from security threats, which can compromise the reliability and admissibility of digital records.
- Rapid IT development and inconsistent security practices across industries exacerbate these risks.
- Courts may impose stringent verification requirements for digital evidence authenticity, though practices vary.
- Digital evidence differs from paper records, requiring closer scrutiny of preservation, authenticity, and relevance to ensure admissibility.
- Effective preservation and security practices are essential to maintain trustworthy and legally acceptable digital evidence.

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Corroborating Digital Evidence



- Digital evidence should be corroborated to strengthen its reliability and evidentiary weight.
- Windows Registry, application logs, and .lnk files can support and explain other evidence but should not be taken at face value.
- Computer clocks are inherently unreliable; discrepancies can affect timelines and event reconstruction.
- Corroboration helps verify authenticity and correctness, reducing the risk of misinterpretation caused by OS or system inconsistencies.
- Documentary evidence (e.g., printed files with microdot identifiers) can support digital records, especially when logs or timestamps are unclear or overwritten.
- Combining multiple sources ensures stronger evidentiary support in legal cases.

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Summary

What was covered today:

- Defined digital evidence and explained its evidentiary value and the conditions it must satisfy to be admissible in legal proceedings
- The special characteristics of digital evidence were described
- The special role of the practitioner to analyze and explain digital evidence to the layperson and the legal fraternity was emphasized
- The reliability of digital evidence in terms of its integrity and use as best evidence was introduced



Reference Source: Practical Digital Forensics (Packt Publishing)

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THANKS – Q&A

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